First-basket variety and 90-day customer behaviour: evidence from UK online retail

Group name	The Outliers
Dataset	UCI Machine Learning Repository: Online Retail (Chen, 2015)
Code	github.com/harshil-0/TheOutliers_DSC (notebook, script and README)

#	Name	Roll no.	Institute email	Contribution / assigned responsibility
1	Aayush Gupta	2023014	aayush23014@iiitd.ac.in	Data provenance and cohort preparation; T1 repurchase inference.
2	Harshil	2023235	harshil23235@iiitd.ac.in	Spending transformation and T2; reproducible notebook and repository.
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5	Jaideep Dahiya	2023254	jaideep23254@iiitd.ac.in	Frequency analysis and T5; report integration and limitations.

AI assistance: Generative AI assisted the project code, statistical analysis and report drafting. The entries above allocate team responsibilities; they are not claims of independent, unaided authorship.

1. Dataset

Source. Chen's *Online Retail* dataset [1], original UCI workbook, retrieved 13 October 2026 (CC BY 4.0). Records span 1 December 2010–9 December 2011. A raw row is an *invoice line*, not a customer or complete basket. The UK retailer sells giftware and serves wholesale customers as well as other buyers.

Raw issues. Missing IDs/descriptions, exact repeats, cancellation invoices, nonpositive quantities/prices and nonmerchandise codes require preparation. Although UCI's catalogue labels missingness "No", the supplied workbook contains missing values; measurements below come from the file.

	Raw	Encoded
Rows	541,909	391,150
Columns	8	60
Rows × columns	4,335,272	23,469,000

Size check: $391{,}150 \times 60 = 23{,}469{,}000 \geq 3{,}000{,}000$. The saved sparse matrix has four numeric features and one-hot indicators for country (37), year-month (13) and observed weekday (6). This is its logical size, not its nonzero count. No customer/invoice/SKU dummy expansion is used. The inferential sample is smaller by design.

Key features

Feature	Type	Real-world meaning
InvoiceNo, CustomerID	Nominal	Invoice and customer linkage keys
StockCode	Nominal	Product/SKU identifier
Quantity	Discrete	Units on an invoice line
UnitPrice	Continuous	Price per unit in GBP
InvoiceDate	Temporal	Transaction timestamp
Country	Nominal	Customer residence country
LineValue	Continuous	Quantity × UnitPrice, GBP
First breadth	Discrete	Distinct SKUs on index day
repeat90	Binary	Any eligible return in days 1–90
spend90	Continuous	Gross future merchandise spend
Return gap	Discrete	Days from index to second day
days90	Discrete	Distinct subsequent shopping days

2. Data preparation

Issue	Action	Reason
Exact duplicates	Remove 5,268; also analyse with repeats retained	Identical lines may be logging repeats or legitimate purchases; check ambiguity.
Cancellation / invalid sale	Remove 11,763 further lines; require non-C invoice, quantity and price > 0	Target gross purchases; do not flip signs or claim net revenue.
Missing customer	Remove 132,186 further lines	Longitudinal linkage needs identity; never impute an ID.
Non-merchandise	Remove 1,542 further lines; code must be five digits plus 0–2 letters	Excludes fees/manual entries; declared heuristic may miss unusual valid SKUs.
Missing description	No imputation; none remain	Variety uses StockCode, not text.
Large orders	Retain all; flag 30,484 lines above GBP 42.075 IQR fence	Wholesale purchases can be real; a tail flag is not proof of error.
Categorical data	One-hot country, month and weekday	Avoid artificial ordinal meaning; save actual encoded matrix.
Same-day invoices	Aggregate by UK customer and date	Split invoices should not count as later visits.

Derived variables. Compute line value, calendar day and distinct daily SKUs. Sum later spend and count distinct shopping days in days 1–90. A non-returner has zero future spend and days, but an unavailable second-basket size and return gap. Those missing quantities are never replaced by zero or an invented return date.

3. What changed after preparation

Measure	Before	After
Transaction rows	541,909	391,150
Raw / encoded columns	8	60
Missing customer IDs	135,080	0
Missing descriptions	1,454	0
Exact duplicate rows	5,268	0
Quantity ≤ 0	10,624	0
UnitPrice ≤ 0	2,517	0
Cancellation rows	9,288	0

Reconciliation: $541{,}909 - 5{,}268 - 11{,}763 - 132{,}186 - 1{,}542 = 391{,}150$. Descriptive issue counts above overlap, whereas these sequential removals do not.

The cleaned table retains eight original fields plus LineValue and Day. Its 4,334 customers and 18,402 invoices include non-UK activity. Restricting to the UK and aggregating gives 15,044 shopping days for 3,916 customers. Cohort eligibility leaves **2,172 customers**; 878 return within 90 days. Identity filtering enables linkage, daily aggregation changes the unit to behaviour, and fixed follow-up makes outcomes comparable. Large transactions remain rather than being deleted to force normality.

4. Basic questions and problem statement

Basic questions

1. How many different products occur in first baskets?
2. What fraction of customers return within 90 days?
3. How uneven is subsequent spending?

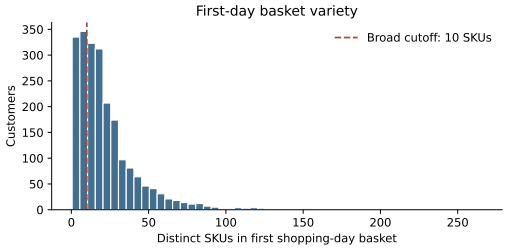


Figure 1: First-day variety is right-skewed (median 17 SKUs). Ten SKUs is a declared segmentation rule, not a fitted optimum.

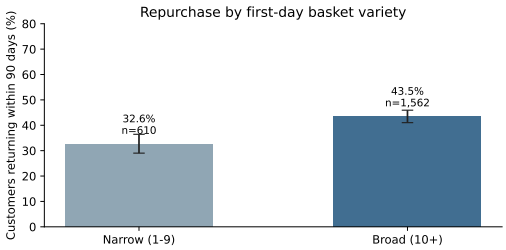


Figure 2: Broad-basket customers return more often: 43.47% versus 32.62%; bars show marginal Wilson 95% intervals.

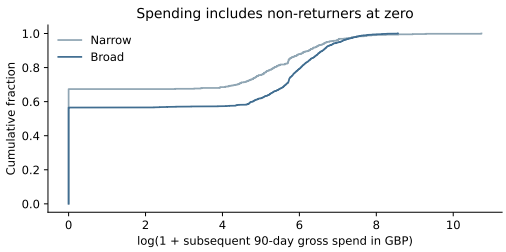


Figure 3: Zero outcomes are common and monetary tails differ. A log-spending contrast is not an arithmetic-revenue gain.

Our problem statements

P1 (primary): Is first-day variety associated with 90-day repurchase and gross spending?

P2: Among returners, does variety change from the first to the second shopping day?

P3: Among returners, does mean return time differ by first-basket breadth?

P4: Among returners, is first-day variety associated with subsequent shopping frequency?

Novelty relative to EDA. No single invoice line contains basket breadth, equal follow-up, or linked first/second behaviour. We construct customer histories to separate occurrence, spending, composition, timing and frequency. This is project-level analytical novelty, not a claim of a previously unstudied topic.

5. Statistical inferences

Unit and cohort. One customer is one analytical record. The index is the earliest eligible UK shopping day in the entire file, restricted to 1 January–31 August 2011. December-observed customers are excluded; this washout does not establish genuine new-customer status. The latest 90-day window ends 29 November, before the file ends. Outcomes exclude the index day. Cancellations/refunds are not matched; spending is gross merchandise purchases, not profit.

Groups. Broad means 10+ first-day SKUs ($n = 1,562$); narrow means 1–9 ($n = 610$). Among 878 returners, group sizes are 679 and 199. Thresholds 5 and 15 are diagnostic checks, not searches for significance.

Analysis history. The supplied plan dated 13 October specifies T1–T3. T4–T5 were specified in a dated 25 October addendum after reviewing the earlier work and before running the added tests. This is transparent staged analysis, not external preregistration or five hypotheses specified before seeing any data.

Error control. All five main tests are two-sided; $\alpha = 0.05$ with Holm adjustment across *all five* p-values. This controls the model-based family-wise error rate without requiring independence between tests [4]. Reported 95% CIs are marginal, not simultaneous. Diagnostic comparisons are not extra confirmatory discoveries. Practical thresholds are explicit analyst choices, not validated commercial standards.

Test 1 (P1): does repurchase probability differ?

a) Hypotheses and design

$\boldsymbol{H_0}$ / $\boldsymbol{H_1}$	$p_B - p_N = 0 \,/\, p_B - p_N \neq 0$
Alpha / side	0.05, five-test Holm / two-sided
Test	Pearson 2×2 chi-square, no continuity correction
Meaningful effect	5 percentage points (pp)

Five pp means 50 additional returning customers per 1,000, a planning threshold rather than established campaign economics.

b) Test choice and alternatives

Two separate groups and a binary response form a contingency table. All expected cells exceed 246; customer aggregation avoids within-customer line pseudoreplication. It does not eliminate shared calendar influences.

Alternative	Comparison / reason
Two-proportion z	Equivalent two-sided test: $z^2 = \chi^2$; not new evidence.
Fisher exact	Valid but unnecessary for large expected counts; $p = 3.68 \times 10^{-6}$.
Yates correction	More conservative; $p = 4.63 \times 10^{-6}$.
McNemar	Requires paired binary observations, absent here.
Adjusted model	Answers a conditional question; exploratory sensitivity below.

c) Result and practical meaning

Group	Return	No return
Broad	679	883
Narrow	199	411

Estimate	$+10.85$ pp (43.47% vs 32.62%)
95% CI	6.32 to 15.22 pp
Statistic (df)	$\chi^2 = 21.432$ (1)
Raw / Holm p	3.67×10^{-6} / 1.10×10^{-5}
Decision	Reject H_0

The Newcombe–Wilson interval for the risk difference lies above 5 pp. This supports potentially useful *unadjusted* segmentation, not the effect of making customers buy more products. Adjustment below weakens evidence of an association beyond basket size.

Test 2 (P1): does subsequent spending differ?

a) Hypotheses and design

Let $Y = \ln(1+S)$, where S is numerical future gross spending in GBP. The currency unit is fixed; adding 1 makes the transformation unit-dependent.

$\boldsymbol{H_0}$ / $\boldsymbol{H_1}$	$\mu_{Y,B} - \mu_{Y,N} = 0 \,/\, \neq 0$
Alpha / side	0.05, five-test Holm / two-sided
Test	Welch independent-samples t
Meaningful effect	$\lvert\Delta_Y\rvert \geq \ln(1.20) = 0.1823$

The threshold corresponds to a 20% geometric-mean ratio of $(1 + S)$, or its reciprocal, not a 20% arithmetic-revenue increase.

b) Test choice and alternatives

Skewed monetary outcomes, independent groups and unknown, potentially unequal variances motivate Welch on the declared log scale. Group log SDs are 3.021 and 2.769. Large samples support approximate inference on means; bootstrap agreement is checked without assuming normal individual purchases.

Alternative	Comparison / reason
Pooled t	Unnecessary equal-variance restriction; $t = 5.115$, $p = 3.41 \times 10^{-7}$. No variance pretest is needed to choose Welch.
Mann–Whitney	Ordering/distribution comparison, not generally a mean or median test; $p = 5.34 \times 10^{-8}$.
Raw-spend Welch	Different, commercially relevant estimand; $p = 0.8481$.
z / paired t	Population variances unknown / no broad–narrow customer pairing.

c) Result and practical meaning

Estimate	0.7210 (2.5947 vs 1.8737)
95% CI	0.4548 to 0.9872
Statistic (df)	$t = 5.314$ (1204.955)
Raw / Holm p	1.28×10^{-7} / 5.12×10^{-7}
Decision	Reject H_0

The exponentiated effect is 2.057 (95% CI 1.576–2.684), exceeding the declared transformed-effect threshold. **Raw mean spending is GBP 248.97 versus GBP 264.45**: the difference is GBP -15.48, CI -174.13 to $+143.17$. There is no clear evidence of higher average revenue in pounds. T2 also mixes return occurrence with spending among returners.

Test 3 (P2): does basket variety change on return?

a) Hypotheses and design

For each returning customer, $D =$ second-day minus first-day distinct SKUs. There are 878 complete pairs.

$\boldsymbol{H_0}$ / $\boldsymbol{H_1}$	$\mu_D = 0$ / $\mu_D \neq 0$
Alpha / side	0.05, five-test Holm / two-sided
Test	Paired t-test
Meaningful effect	$\lvert\mu_D\rvert \geq 2$ SKUs

Two SKUs is an interpretable composition threshold; it does not establish profitability.

b) Test choice and alternatives

The same customers provide both baskets. Inference uses their within-customer differences (SD 20.234; skewness -0.617), not separate normality assumptions for each basket. With $n = 878$, the paired-mean approximation is supported by the bootstrap.

Alternative	Comparison / reason
Independent t	Discards the pairing and misstates the standard error.
Wilcoxon signed rank	Different location target with shape assumptions; $p = 5.64 \times 10^{-27}$, 53 zero differences omitted.
Sign test	Uses direction, ignores magnitude; $p = 3.97 \times 10^{-24}$.
Customer bootstrap	Resamples complete paired differences; CI below.

c) Result and practical meaning

Estimate	-6.092 SKUs (24.841 to 18.748)
95% CI	-7.432 to -4.752 SKUs
Statistic (df)	$t = -8.922$ (877)
Raw / Holm p	2.63×10^{-18} / 1.32×10^{-17}
Decision	Reject H_0

The interval lies beyond two SKUs in absolute magnitude. Observed returners buy fewer distinct products on their second day on average. Initial stocking or regression to the mean could contribute; this does not establish declining loyalty or describe non-returners.

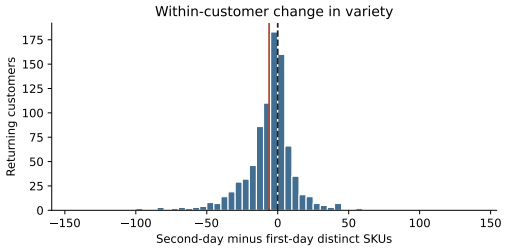


Figure 4: Paired changes tend negative, but individual changes vary widely. All observed differences are included.

Test 4 (P3): does return timing differ?

a) Hypotheses and design

Among returners only, let $G =$ days between index and second shopping day (range 1–90). Broad and narrow samples have 679 and 199 customers. Non-returners do not receive an imputed return gap.

$\boldsymbol{H_0}$ / $\boldsymbol{H_1}$	$\mu_{G,B} - \mu_{G,N} = 0$ / $\neq 0$
Alpha / side	0.05, five-test Holm / two-sided
Test	Welch independent-samples t
Meaningful effect	Absolute difference of 7 days

Seven days represents one weekly contact-planning cycle, an analyst-selected threshold rather than validated marketing economics.

b) Test choice and alternatives

The groups consist of different customers. A gap computed from two dates does not make the group comparison paired. Welch targets mean calendar days without equal-variance assumptions. The bounded outcome and sample sizes support approximate mean inference; customer bootstrap checks it.

Alternative	Comparison / reason
Mann–Whitney	Ordering/distribution sensitivity, not the same mean-gap null; $U = 68,735.5$, $p = 0.7089$.
Pooled / paired t	Adds equal-variance assumption / no paired broad–narrow customers.
Survival analysis	Appropriate for time-to-return in the *whole cohort*, with non-returners censored. Our conditional question is different.
Bootstrap	5,000 within-group customer resamples, seed 5584; mean-gap CI below.

c) Result and practical meaning

Group means (SD)	Broad 43.318 (24.402); narrow 42.548 (24.210) days
Median [Q1, Q3]	Broad 42 [21.5, 63]; narrow 41 [23, 62.5] days
Mean gap / 95% CI	0.770 days / -3.076 to 4.617
Statistic (df)	$t = 0.394$ (325.052)
Raw / Holm p	0.6938 / 0.9692
Decision	Do not reject H_0

We find no clear mean-timing difference among observed returners. The marginal interval includes zero and lies within ± 7 days, suggesting any difference compatible with this interval is smaller than the proposed weekly threshold. This is not a prespecified equivalence test, a simultaneous bound, or a statement about the whole cohort.

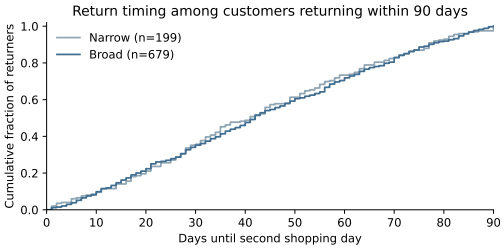


Figure 5: Return-gap distributions largely overlap among observed returners; excluded non-returners are not assigned fictitious times.

Test 5 (P4): does variety track purchase frequency?

a) Hypotheses and design

Among the 878 returners, $X =$ first-day SKU count and $F =$ subsequent shopping days in the original days-1–90 window. The initial day is excluded; the first return is included. Use numeric breadth, not the broad/narrow split.

$\boldsymbol{H_0}$ **/** $\boldsymbol{H_1}$	$\rho_s(X, F \mid \text{return90}) = 0 / \neq 0$
Alpha / side	0.05, five-test Holm / two-sided
Test	Spearman rank correlation; average ranks for ties
Magnitude of interest	$\lvert\rho_s\rvert \geq 0.10$

The 0.10 threshold is an analyst-chosen modest association of interest; it is not a universal commercial cutoff or a 10% increase in visits. Of the returners, 549 have one later shopping day and 329 have two or more (range 1–15; mean 1.687).

b) Test choice and alternatives

Spearman targets monotonic association between skewed discrete quantities, without requiring a straight-line relationship or normal marginals. Ties are retained. The p-value uses a large-sample approximation ($n = 878$), not an exact no-ties calculation [2]. Zero rank correlation does not imply independence or exclude a non-monotonic relationship.

Alternative	Comparison / reason
Pearson correlation	Targets linear raw-scale association; more sensitive to extreme magnitudes.
Kendall tau-b	Tie-aware diagnostic; $\tau_b = 0.0185$, $p = 0.4837$. Different effect scale.
Chi-square	Requires unnecessary categorical cutoffs for these counts.
Ordinary Poisson	Needs count-model assumptions and would ignore zero truncation from selecting returners.
Bootstrap CI	Resample 5,000 complete customer pairs, rerank each resample; seed 5585.

c) Result and practical meaning

Estimate / statistic	$\rho_s = 0.0236$; $n = 878$
95% bootstrap CI	-0.0413 to 0.0890
Approximate t (df)	0.699 (876)
Raw / Holm p	0.4846 / 0.9692
Decision	Do not reject H_0

The reference statistic is $t = \rho_s\sqrt{(n-2)/(1-\rho_s^2)}$; df $= n - 2$ belongs to this approximation, not an exact rank distribution. The interval crosses zero and stays inside ± 0.10. Evidence of a meaningful monotonic frequency association is absent under the stated model and threshold; this is not proof of no relationship or a formal equivalence test.

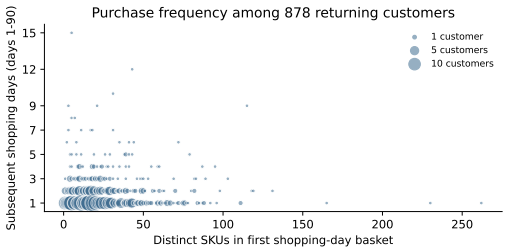


Figure 6: Many returners buy on only one later day. Marker area reflects the number of customers at each observed breadth/frequency combination.

Robustness and synthesis

Bootstrap. T1–T2 resample customers separately within groups; T3 resamples paired differences (5,000 draws, seed 558). T4 uses seed 5584, and T5 resamples both measurements together with seed 5585. Percentile intervals are approximate and do not remove confounding [3].

Effect	Main 95% CI	Bootstrap 95% CI
T1 (pp)	6.32 to 15.22	6.347 to 15.286
T2 (log)	0.4548 to 0.9872	0.446 to 0.985
T3 (SKUs)	-7.432 to -4.752	-7.383 to -4.771
T4 (days)	-3.076 to 4.617	-3.055 to 4.549

Analytical and bootstrap intervals agree closely. T5's main interval is already the customer-pair bootstrap interval.

Duplicate sensitivity. Keeping exact repeated lines leaves T1, T3, T4 and T5 unchanged. T2's log gap becomes 0.7219 (main 0.7210), five-test Holm $p = 5.02 \times 10^{-7}$. The substantive conclusions persist.

Cutoff sensitivity. All variants use the same cohort and five-test Holm families; these checks are diagnostic.

Cutoff	T1 gap (pp); Holm p	T2 log gap	T4 days; Holm p
5+	8.02; 0.0441	0.5820	2.936; 0.6094
10+	10.85; 1.10×10^{-5}	0.7210	0.770; 0.9692
15+	10.66; 1.62×10^{-6}	0.7134	1.065; 0.9692

T1–T2 directions persist and T4 remains nonsignificant. T3 and T5 do not use the breadth split. This does not establish an optimal cutoff; T4's 5-SKU sensitivity CI (-2.710 to 8.583 days) also shows that the weekly practical bound is not uniform across choices.

Adjustment for initial basket size. An exploratory linear-probability model includes broad status, log initial spend, log initial units and index-month indicators (January reference), using all 2,172 customers. The design has 11 full-rank columns; HC3 heteroskedasticity-robust errors give a broad coefficient of $+4.98$ pp, CI -0.19 to $+10.15$ pp, $z = 1.889$, $p = 0.0588$. Fitted probabilities remain in [0,1]. Initial-spend medians differ: GBP 315.84 versus 138.28. Incremental value beyond basket size remains uncertain.

Integrated answer. First-basket breadth is an unadjusted marker of returning and higher log-transformed future spending. It does not establish higher arithmetic revenue. Among returners, basket variety declines, while the added tests do not show faster return or a meaningful monotonic frequency association. Timing and frequency are related: an earlier return leaves more time for further visits within the same window. These are complementary descriptions, not independent causal mechanisms.

6. Limitations

- **Association and confounding.** Customer type, wholesale activity, acquisition channels and product mix may influence both basket breadth and future purchases. Adjustment covers only measured proxies and is not causal identification.
- **History and selection.** First observed is not first-ever purchase. Unknown IDs, non-UK activity and later cohorts are excluded. Results may not generalise to those populations, other retailers or current markets.
- **Gross purchases.** Refunds and cancellations are not matched. We cannot infer net revenue, profit or intervention return on investment. The merchandise-code heuristic may omit genuine products.
- **Conditional returner analyses.** T3–T5 select 878 observed returners. Selection can induce associations; none describes an unobserved second basket or return time for non-returners. T3 can reflect stocking or regression to the mean. T5 is a count in the original window, not a post-return daily rate.
- **Dependence and seasonality.** Customers may share promotions or calendar shocks. Equal follow-up and customer-level resampling avoid line-level pseudoreplication but do not fully model common influences.
- **Model and threshold uncertainty.** Ten SKUs, 5 pp, the 20% transformed ratio, two SKUs, seven days and $|\rho_s| = 0.10$ are declared analytical choices. Inference is approximate and CIs are marginal. Non-rejection does not prove equality; no formal equivalence tests were specified.
- **Analysis history and sampling.** T4–T5 are disclosed later additions, not untouched confirmatory hypotheses. Holm corrects the stated family, not hypothetical unreported selection. This is a historical available-record cohort, not a probability sample of all retail customers.

References and reproducibility

[1] Chen, D. (2015). *Online Retail* [Dataset]. UCI. doi:10.24432/C5BW33. CC BY 4.0.

[2] SciPy documentation: Welch t, paired t, chi2_contingency, Spearman, Kendall tau-b.

[3] SciPy: bootstrap resampling and paired observations.

[4] R statistical documentation: Holm p-value adjustment; Holm (1979), *Scandinavian Journal of Statistics*, 6, 65–70.

The linked repository supplies the executed notebook, script, plan, dictionary, raw/processed data, all results, figures and report source. Run `python analysis.py`; the notebook contains the same visible analysis. Versions and source SHA-256 are recorded. Report numerical tables are generated from `results.json` to avoid stale values. All five main tests and declared sensitivities are reported.